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Project Title:	Role of Growth Hormone Secretogoues and Exercise on Muscle Homeostasis Under Microgravity		
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	We propose to test the hypothesis that increased systemic levels of growth hormone (GH) through stimulated growth hormone releasing hormone (GHRH) action on the anterior pituitary in concert with resistive exercises may provide a long term solution for muscle homeostasis under conditions of prolonged weightlessness. As a key hormone replacement therapy, induced secretion of GH, by GH secretogoues will be studied for its potential on muscle mass in small mammals under simulated microgravity. Induction of GH would provide a chronic pulsatile source of a growth-promoting protein whose biosynthesis or secretion is reduced in long term spaceflight. Mice will be used in hind-limb suspension models to determine the role of a novel drug MK-0677 on maintenance of muscle mass and whether concentric exercises might act in synergy with hormone treatment. As a means to engineer and ensure long-term protein production that would be workable in humans, gene therapy technology will also be employed to monitor muscle mass preservation during hindlimb suspension, after direct intramuscular injection of a genetically engineered		

Task Description:	muscle-specific vectors expressing GHRH. A current goal of the NASA-HEDS2 is to develop countermeasures that allow humans to live and work in microgravity for duration over a year and to minimize readapting to Earth's gravity, and optimize crew safety, well-being, and performance. We propose to test the hypothesis that the GH/IGF-I axis through autocrine/paracrine mechanisms may provide long term muscle homeostasis under conditions of prolonged weightlessness. As a key alternative to hormone replacement therapy, ectopic production of hGH, growth hormone releasing hormone (GHRH), will be studied for its potential on muscle mass impact in transgenic mice under simulated microgravity. Transgenic mice will be used in hind-limb suspension models to determine the role of GH/IGF-I on maintenance of muscle mass and whether concentric exercises might act in synergy with hormone treatment. As a means to engineer and ensure long-term protein production that would be workable in humans, gene therapy technology will be used by to monitor muscle mass preservation during hind-limb suspension, after direct intramuscular injection of a genetically engineered muscle-specific vector expressing GHRH. Effects of this gene-based therapy will be assessed in both fast twitch (medial gastrocnemius) and slow twitch muscle (soleus). End-points include muscle size, ultrastructure, fiber type, and contractile function, in normal animals, hind limb suspension, and reambulation. The following aims are: The following aims are: Specific Aim I: To determine the role of overexpression of hGH in transgenic mice on muscle mass accretion under condition of hindlimb under loading. Specific Aim II: Does Growth hormone secretagogue, MK-0677, provide signaling for muscle mass homeostasis in response to unloading? Specific Aim III: Does ectopic expression of GHRH via injectable muscle expression vectors and MK- 0677 act in synergy to potentiate the secretion of GH? Effects of this gene-based therapy will be assessed together with MK-0677 to asses their synergy. End-points include muscle size, ultrastructure, fiber type, and contractile function, in both fast twitch (medial gastrocnemius) and slow twitch muscle (soleus) muscle of normal animals, hindlimb suspension, and reambulation. Answering these aims will provide information that could: a) refine training programs for astronauts, b) provide the basis for the development of pharmaceutical countermeasures, and c) provide marker(s) to assess the success of exercise countermeasures in the prevention of muscle atrophy in astronauts. Specific Hypotheses 1. GHRH/GH/IGF-I axis through autocrine and paracrine mechanisms will provide long term muscle hypertrophy under conditions of prolonged weightlessness or muscle unweighting due to suspension.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our study will provide a framework against which other types of therapies can be compared, both in clinical studies on GH deficient children or in elderly, where our therapy could be combined with an adequate diet for optimized results in growth, muscle strength and/or bone mineralization. As a key hormone replacement therapy, induced secretion of GH by GH secretogoues, has been studied for its potential on improving nutrition, skeletal growth and maintenance of muscle homeostatsis. We will continue upon our proposed studies to determine if the upregulation of the GHRH/GH/IGF-I axis will act in synergy with resistive exercise regimens to block muscle atrophy under microgravity. Enhancement in muscle mass could occur by GHRH secretagogue activity increasing the systemic levels of growth hormone, which then elicits increased level of systemic or local IGF-I production in muscle. It is very important to note that space flight conditions induce stress to some degree by heavy work loads, lack of sleep, impaired circadian daylight cycles and nutrition. Hormones secreted in greater amounts as part of the stress response include ACTH and cortisol which increased greater than 100% during Skylab missions Cortisol is one of the primary muscle catabolic factors that rapidly elicits muscle protein break down and impairs GH secretion and reduces IGF-I levels, increases bone loss and damages the immune system. Also, when adult rats were treated with dexamethasone, IGF-I was still able to stimulate protein synthesis in the epitrochlearis muscle, but only by 41.9% of the increase found in a pair-fed control group (Dardevet et al. J. Endocr.. 156:83, 1998). As hindlimb unloaded mice should have increases in plasma glucocorticoids, it is possible that this would suppress the action of IGF-I. Furthermore, Dr. Fred Goldberg showed three decades ago that inactive skeletal muscle becomes more sensitive to glucocorticoids (J. Physiol. 200:667, 1969).). Thus, we will also plan to test the myogenic injectable GHRH vector system in rats treated with glucocorticoids as a therapy to block steroid induced muscle catabolism. Clearly, development of the injectable GHRH gene therapy has great synergy with the goals set forth for human countermeasures.. Although, It will be up to the NASA leadership to decide if human gene therapy is a logical medical treatment for long duration space flight, we also believe that Alzet like slow release pumps releasing protease resistant GHRH might provide acceptable long term benefit to the astronauts to ameliorate muscle and bone atrophy.
Task Progress:	IMPLICATIONS OF PROJECT FINDINGS FOE FUTURE RESEARCH Our study have provided a framework against which other types of therapies can be compared, both in clinical studies on GH deficient children or in elderly, where our therapy could be combined with an adequate diet for optimized results in growth, muscle strength and/or bone mineralization. As a key hormone replacement therapy, induced secretion of GH by GH secretogoues, has been studied for its potential on improving nutrition, skeletal growth and maintenance of muscle homeostatsis. We will continue upon our proposed studies to determine if the upregulation of the GHRH/GH/IGF-I axis will act in synergy with resistive exercise regimens to block muscle atrophy under microgravity. Enhancement in muscle mass could occur by GHRH secretagogue activity increasing the systemic levels of growth hormone, which then elicits increased level of systemic or local IGF-I production in muscle. It is very important to note that space flight conditions induce stress to somedegree by heavy work loads, lack of sleep, impaired circadian daylight cycles and nutrition. Hormones secreted in greater amounts as part of the stress response include ACTH and cortisol which increased greater than 100% during Skylab missions Cortisol is one of the primary muscle catabolic factors that rapidly elicits muscle protein break down and impairs GH secretion and reduces IGF-I levels, increases bone loss and damages the immune system. Also, when adult rats were treated with dexamethasone, IGF-I was still able to stimulate protein synthesis in the epitrochlearis muscle, but only by 41.9% of the increase found in a pair-fed control group (Dardevet et al. J. Endocr.. 156:83, 1998). As hindlimb unloaded mice should have increases in plasma glucocorticoids, it is possible that this would suppress the action of IGF-I. Furthermore, Dr. Fred Goldberg showed three decades ago that inactive skeletal muscle becomes more sensitive to glucocorticoids (J. Physiol. 200:667, 1969).).

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